

Off-Diagonal Inverse

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1 Problem Statement

Given a model with N parameters $\theta(t) = \{\theta^i(t)\}_{i=1}^N$ and a loss function $\mathcal{L} : \mathbb{R}^N \rightarrow \mathbb{R}$, our goal is to find the most optimal parameter update $\delta\theta$ at each time step t . That is, in gradient descent we have

$$\frac{d\theta^i}{dt} = -g^{ij} \frac{\partial \mathcal{L}}{\partial \theta^j} \implies \delta\theta^i = -\eta g^{ij} \frac{\partial \mathcal{L}}{\partial \theta^j}$$

where $\eta \in \mathbb{R}^{>0}$ is the learning rate and g^{ij} is the inverse of the induced metric on the parameter space.

2 Review of Diagonal Metric Update

The original ambient metric in the paper is

$$h = \begin{pmatrix} \gamma & \vec{0} \\ \vec{0}^\top & 1 \end{pmatrix} \in \mathbb{R}^{(N+1) \times (N+1)}$$

where $\gamma \in \mathbb{R}^{N \times N}$. What we really care about is the induced metric on the parameter space, which is given by the pullback of h under the embedding $\phi : \mathbb{R}^N \rightarrow \mathbb{R}^{N+1}$ defined by $\phi(\theta^1, \dots, \theta^N) = (\theta^1, \dots, \theta^N, f(\theta^1, \dots, \theta^N))$ where $f : \mathbb{R}^N \rightarrow \mathbb{R}$ is some scalar function which we take to be the loss function $\mathcal{L}$. The induced metric g on the parameter space is given by

$$g_{ij} = h_{\alpha\beta} \frac{\partial \phi^\alpha}{\partial \theta^i} \frac{\partial \phi^\beta}{\partial \theta^j} = \gamma_{ij} + \frac{\partial \mathcal{L}}{\partial \theta^i} \frac{\partial \mathcal{L}}{\partial \theta^j}.$$

Then, to calculate the parameter updates $\delta\theta^i$ we need g^{ij} , the inverse of g_{ij} . We can compute this inverse using the Sherman-Morrison-Woodbury formula:

Sherman-Morrison-Woodbury Formula

Let $A \in \mathbb{R}^{n \times n}$ be an invertible matrix, and let $U, V \in \mathbb{R}^{n \times k}$ be matrices. Then, if $C = A + UV^\top$ is invertible, we have

$$C^{-1} = A^{-1} - A^{-1}U(I_k + V^\top A^{-1}U)^{-1}V^\top A^{-1}.$$

This gives us

$$g^{ij} = \gamma^{ij} - \frac{\gamma^{ik}\gamma^{jl} \frac{\partial \mathcal{L}}{\partial \theta^k} \frac{\partial \mathcal{L}}{\partial \theta^l}}{1 + \gamma^{mn} \frac{\partial \mathcal{L}}{\partial \theta^m} \frac{\partial \mathcal{L}}{\partial \theta^n}} = \gamma^{ij} - \frac{l^i l^j}{1 + l_k l^k}$$

where

$$l_i := \frac{\partial \mathcal{L}}{\partial \theta^i} \implies l^i = \gamma^{ij} l_j.$$

Thus, the parameter updates are given by

$$\begin{aligned} \delta \theta^i &= -\eta \left(\gamma^{ij} - \frac{l^i l^j}{1 + l_k l^k} \right) l_j \\ &= -\eta l^i \left(1 - \frac{l_j l^j}{1 + l_k l^k} \right) \\ &= -\eta l^i \left(\frac{1}{1 + l_k l^k} \right). \end{aligned}$$

So to summarize, (1) we start with an ambient metric h which induces a metric g on the parameter space, (2) we compute the inverse g^{ij} , and (3) we use g^{ij} to compute the parameter updates $\delta \theta^i$.

3 Off-Diagonal Metric Update

Now, we consider a more general ambient metric of the form

$$h = \begin{pmatrix} \gamma & \vec{a} \\ \vec{b}^\top & 1 \end{pmatrix} \in \mathbb{R}^{(N+1) \times (N+1)}$$

where $\vec{a}, \vec{b} \in \mathbb{R}^N$. Intuitively, the vectors $\vec{a}$ and $\vec{b}$ introduce off-diagonal couplings between the parameter space and the loss function dimension. The induced metric on the parameter space is then given by

$$\begin{aligned} g_{ij} &= h_{\alpha\beta} \frac{\partial \phi^\alpha}{\partial \theta^i} \frac{\partial \phi^\beta}{\partial \theta^j} \\ &= \gamma_{ij} + a_i \frac{\partial \mathcal{L}}{\partial \theta^j} + b_j \frac{\partial \mathcal{L}}{\partial \theta^i} + \frac{\partial \mathcal{L}}{\partial \theta^i} \frac{\partial \mathcal{L}}{\partial \theta^j} \\ &= \gamma_{ij} + a_i l_j + b_j l_i + l_i l_j. \end{aligned}$$

To compute the inverse g^{ij} , we notice that g can be expressed as a rank-2 update to γ , that is for $U = [a, l], V = [l, b + l] \in \mathbb{R}^{N \times 2}$ we have

$$\begin{aligned}\gamma_{ij} + U_{ik}V_{jk} &= \gamma_{ij} + a_i l_j + (b_j + l_j) l_i \\ &= \gamma_{ij} + a_i l_j + b_j l_i + l_i l_j \\ &= g_{ij}\end{aligned}$$

and so we can apply the Sherman-Morrison-Woodbury formula again to get

$$\begin{aligned}g^{ij} &= \gamma^{ij} - \gamma^{ik} U_{k\alpha} (\delta_{\alpha\beta} + V_{\beta l} \gamma^{lm} U_{m\alpha})^{-1} V_{\beta n} \gamma^{nj} \\ &= \gamma^{ij} - \gamma^{ik} (a_k, l_k) \begin{pmatrix} 1 + (b_m + l_m) \gamma^{mn} a_n & (b_m + l_m) \gamma^{mn} l_n \\ l_m \gamma^{mn} a_n & 1 + l_m \gamma^{mn} l_n \end{pmatrix}^{-1} \begin{pmatrix} b_n + l_n \\ l_n \end{pmatrix} \gamma^{nj} \\ &= \gamma^{ij} - \gamma^{ik} (a_k, l_k) \begin{pmatrix} 1 + b_n a^n + l_n a^n & b_n l^n + l_n l^n \\ l_n a^n & 1 + l_n l^n \end{pmatrix}^{-1} \begin{pmatrix} l_n \\ b_n + l_n \end{pmatrix} \gamma^{nj} \\ &= \gamma^{ij} - \gamma^{ik} \frac{1}{D} (a_k, l_k) \begin{pmatrix} 1 + l_n l^n & -(b_n l^n + l_n l^n) \\ -l_n a^n & 1 + b_n a^n + l_n a^n \end{pmatrix} \begin{pmatrix} l_n \\ b_n + l_n \end{pmatrix} \gamma^{nj}\end{aligned}$$

where $D = (1 + b_n a^n + l_n a^n)(1 + l_n l^n) - (b_n l^n + l_n l^n)(l_n a^n)$.

Quick Sanity Check

If we set $\vec{a} = \vec{0}$ and $\vec{b} = \vec{0}$, we recover the diagonal metric case:

$$\begin{aligned}g^{ij} &= \gamma^{ij} - \gamma^{ik} \frac{1}{1 + l_n l^n} (0, l_k) \begin{pmatrix} 1 + l_n l^n & -l_n l^n \\ 0 & 1 \end{pmatrix} \begin{pmatrix} l_n \\ l_n \end{pmatrix} \gamma^{nj} \\ &= \gamma^{ij} - \gamma^{ik} \frac{1}{1 + l_n l^n} (0, l_k) \begin{pmatrix} (1 + l_n l^n) l_n - (l_n l^n) l_n \\ l_n \end{pmatrix} \gamma^{nj} \\ &= \gamma^{ij} - \gamma^{ik} \frac{1}{1 + l_n l^n} (0, l_k) \begin{pmatrix} l_n \\ l_n \end{pmatrix} \gamma^{nj} \\ &= \gamma^{ij} - \frac{\gamma^{ik} l_k l_n \gamma^{nj}}{1 + l_n l^n} \\ &= \gamma^{ij} - \frac{l^i l^j}{1 + l_k l^k}\end{aligned}$$

which matches our previous result.

Thus, the parameter updates are given by

$$\begin{aligned}
\delta\theta^i &= -\eta g^{ij} l_j \\
&= -\eta \left[\gamma^{ij} - \gamma^{ik} \frac{1}{D} (a_k, l_k) \begin{pmatrix} 1 + l_n l^n & -(b_n l^n + l_n l^n) \\ -l_n a^n & 1 + b_n a^n + l_n a^n \end{pmatrix} \begin{pmatrix} l_n \\ b_n + l_n \end{pmatrix} \right] l_j \\
&= -\eta \left[l^i - \gamma^{ik} \frac{1}{D} (a_k, l_k) \begin{pmatrix} 1 + l_n l^n & -(b_n l^n + l_n l^n) \\ -l_n a^n & 1 + b_n a^n + l_n a^n \end{pmatrix} \begin{pmatrix} l_n \\ b_n + l_n \end{pmatrix} \right] \\
&= -\eta \left[l^i - \gamma^{ik} \frac{1}{D} (a_k [(1 + l_n l^n) l_n - (b_n l^n + l_n l^n)(b_n + l_n)] + l_k [-l_n a^n l_n + (1 + b_n a^n + l_n a^n)(b_n + l_n)]) \right] \\
&= -\eta \left[l^i - \frac{1}{D} (a^i [(1 + l_n l^n) l_n - (b_n l^n + l_n l^n)(b_n + l_n)] + l^i [-l_n a^n l_n + (1 + b_n a^n + l_n a^n)(b_n + l_n)]) \right]
\end{aligned}$$

4 Results